Andrew Nguyen

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Education

University of Michigan | **Aug. 2023 – May 2027**

B.S. in Computer Science & B.S. in Robotics, Entr. Minor, Eng. Honors Program, GPA: 3.8/4.0 | *Ann Arbor, MI*

Coursework: Machine Learning, Bayesian ML, Algorithms for Robotics, Robot Operating Systems, SLAM

Experience

ARCAD Lab (Project with Schaeffler) | **Sep. 2025 – Present**

Robotics Simulation Engineer — Python, MuJoCo, NumPy, PyTorch, RL, ROS, Git | *Ann Arbor, MI*

- Integrating actuator dynamics and RL locomotion policy into MuJoCo humanoid sim to bridge Sim to Real gap; validating fidelity using hardware dyno data
- Developing hybrid actuator models by synthesizing analytical physics-based equations with Neural Networks to capture complex non-linear behaviors

FREE Lab | **Aug. 2023 – Present**

Undergraduate Research Assistant — Python, PyBullet, Git, Onshape | *Ann Arbor, MI*

- Modeling interconnected fixed-wing UAVs for increased efficiency and maneuverability using PyBullet
- Designed and prototyped passive wall-climbing robot using bi-stable suction; successfully scaled vertical plaster & glass surfaces carrying 8kg payload (first-author paper accepted to ICRA 2026)
- US Provisional Patent App. No. 63/903,627, "Wall-Climbing Robot," Filed 2025
- *Outstanding Presentation Award*, Gulf Coast Undergraduate Research Symposium, Rice University, November 2024

RAI Institute (Formerly Boston Dynamics AI Institute) | **May – Aug. 2024 & May – Aug. 2025**

Hardware Intern — Python, Git, Onshape, Linux | *Cambridge, MA*

- Executed several critical-path tasks for a project to advance state-of-the-art robotics performance
- Wrote automated motor dynamometer test suite in Python to validate actuator builds
- Analyzed coulomb friction, viscous, and torque-velocity plots of custom actuators, adjusting assembly as necessary
- Assembled 40 quasi-direct-drive actuators using personally designed test fixtures (Onshape, CNC, metal lathe)

The Brainstormers FTC Team 8644 | **2012 – 2023**

Team Captain, Head Robot Designer, Software Lead — Java, Git, Fusion 360, OpenCV | *Lexington, MA*

- Led engineering and software for competitive robotics team; wrote 4,000+ lines of Java control and OpenCV vision
- 2nd Place World Championship (2022), 6x State Champion; robot ranked #1 globally (2023), World Champion (2018)
- Mentored 100+ teams and presented work to MA Lt. Governor and corporate partners

Build-It-Yourself | **May 2020 – Dec. 2022**

Software Engineering Intern — JavaScript, HTML/CSS, Git | *Cambridge, MA*

- Member of Invention Universe software team, worked with college students on beta for over 800 users
- Presented company business model at MIT Universal Village 2022 Conference
- Independently developed new front-end features in JavaScript: likes, comments, and Hall-of-Fame which are deployed
- Responsible for Usage Analytics tracking for IU and regularly updated the team on consumer metrics
- Joined Business Development team in June 2022 to help spread BIY to a wider audience globally

Publications

- Andrew Nguyen, Mingyuan Li, Daniel Bruder, "A High-Payload Wall-Climbing Robot Using Passive Bistable Suction Cups", Accepted for publication, 2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria

Projects

Chunk (It)! | **Mar. 2024 – May 2024**

- Developed a website in JavaScript to help students manage time and calculate task completion time
- Won $3,000 and placed 3rd in Bobcat Ventures Pitch Competition of 11 ideas at Bates College

Technical Skills

Languages: C++, Python, ROS, C, Java, HTML/CSS, JavaScript, Julia
Libraries: NumPy, Pandas, Matplotlib, PyTorch, MuJoCo, PyBullet
Fabrication: 3D Printing, Machine Shop Tools, CNC Routing, Metal Lathe, Silicon Molding
Applications: Onshape, SolidWorks, Autodesk Fusion 360, 3D Printer Slicers
Tools: Git, Linux, CI/CD, Unit Testing, Jupyter, OpenCV